Bryan Torres

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) **Blacksburg, Virginia**
Master of Engineering in Computer Science & Applications Expected May 2027
Graduate Coursework GPA: 3.90 / 4.00 | **Focus:** Computer Systems & Architecture, Software Systems
Bachelor of Science in Computer Science Aug. 2022 – May 2026

EXPERIENCE

Software Engineer Intern | **Expedition Technology** **May 2026 – Aug. 2026**
Geospatial Dataset Diversity *Herndon, Virginia*

- Built **Python/FastAPI APIs** to score geospatial dataset diversity using normalized Shannon entropy across geographic, temporal, illumination, and image-quality metadata; developed with mock data and validated in a secure environment.
- Integrated APIs with a **TypeScript/Vite** frontend and built Chart.js and GeoJSON/TopoJSON visualizations to identify dataset coverage gaps and distribution patterns.
- Automated pre-deployment **database migrations** for Kubernetes services using reusable **Helm/Argo CD** templates and sync waves to sequence secrets, migrations, and application deployment, blocking releases when migrations fail.

Software & DevOps Engineer Intern | **Dark Wolf Solutions** **Jun. 2025 – Aug. 2025**
Project Syndicate *Herndon, Virginia*

- Built and deployed a CI/CD security platform with **four containerized services** (Next.js, FastAPI, PostgreSQL, Dex/OIDC) on **Google Kubernetes Engine** using Docker, Kubernetes, and Argo CD.
- Implemented **GitLab CI security gates** with SonarQube and OWASP checks; built Jest/pytest suites reaching **80% test coverage** and reduced container image size by **22%**.

Undergraduate Researcher | **Virginia Tech** **Oct. 2024 – Jan. 2026**
MARCO – Multi-Agent Code Optimization for HPC *Blacksburg, Virginia*

- Co-developed a **Python orchestration pipeline** for LLM-generated CUDA kernels, automating compilation, execution, and output verification with a bounded, diagnostic-driven repair loop.
- Built an **evaluation harness** benchmarking MARCO against HPC-Coder, automating **Slurm-scheduled GPU runs**, parameterized GEMM sweeps, and Nsight profiling, and showing MARCO outperforming the baselines.
- **Publication:** Co-authored *MARCO: Multi-Agent Code Optimization with Real-Time Knowledge Integration for High-Performance Computing* [arXiv, 2025].

PROJECTS

Kingdom Tycoon | **Roblox Game (In Development)** **Sep. 2026 – Present**
Luau, Roblox Studio, DataStoreService, ProfileStore *Blacksburg, Virginia*

- Architected a **modular Luau backend** with separate server services, client controllers, shared types, and explicit initialization lifecycles.
- Implemented **persistent player profiles** for currency, progression, and building purchases using DataStoreService and ProfileStore, integrating session locking, autosaving, and disconnect/shutdown saves; verified persistence across play sessions.
- Built **snapshot-based state updates** with schema validation and revision checks to reject invalid or stale mutations; added versioned schemas and tests for data isolation and invalid-update rejection.

LEADERSHIP

Mentor | **CS Careers at Virginia Tech** **Sep. 2026 – Present**
Peer Mentorship & Career Development *Blacksburg, Virginia*

- Mentor **3 CS underclassmen** through resume and project reviews, coursework support, and career guidance.

TECHNICAL SKILLS

- **Languages:** Python, C, Luau, TypeScript, JavaScript, SQL, Java
- **Frameworks & Data:** React, FastAPI, Next.js, Vite, Node.js, REST APIs, PostgreSQL, SQLAlchemy, Alembic
- **Cloud & Tools:** Docker, Kubernetes, Helm, Argo CD, GitLab CI/CD, Google Cloud Platform, Git/GitHub, Linux
- **Testing & AI:** pytest, Jest, Unit/Integration Testing, LLMs, OpenAI/Anthropic APIs